MIS 201 · WEEK 5 · MONDAY 10/19

Ch. 5: The Machine Has a Body

Part 1. Chapter 5: every scroll, stream, and prompt runs in a building with a power bill, a water bill, and neighbors. Plus the case: AI's hunt for power.

Part 2. A class discussion, a break, then project teams.

Part 3. Midterm review for Wednesday.

Today

PART 1 · CHAPTER 5 AND THE CASE · ~40 MIN

Scroll first: where does your feed live?	~5 min
The building, and what AI changes	~7 min
Two costs, and how your habits compare	~8 min
Who pays, and who's counting	~5 min
The case: Three Mile Island to Memphis	~8 min
Rules, and your decision	~5 min
Lab 5 this week	~2 min

DISCUSSION ~15 · BREAK ~5 · TEAMS ~20 · REVIEW ~30

Discussion: three questions, then the whole class	~15 min
Break, then find your project group	~25 min
Midterm review: the debates again, plus sample questions	~30 min

WHY CHAPTER 5 TODAY?

Not on Wednesday. Still worth it.

It isn't on the midterm. It is on the final, and Lab 5 runs on it.

Lab 5 is at home this week, due Friday 10/23 with the Chapter 5 quiz. After today you can do it any night, not Friday at 11 p.m.

PART 1 · CHAPTER 5 · ABOUT 40 MINUTES

Before anything else, scroll on something.

Phones out. Open TikTok, Reels, Shorts, whatever you scroll. Two minutes. Then put it face down and answer two questions with the person next to you:

1. Where did those videos come from?

2. What did it take to recommend each one for you?

Chapter 5 in one sentence: every AI system runs on infrastructure, the same buildings run everything else we do online, and AI's cost is small per use, large in total, and hard to be exact.

Our feeds are AI, and they live in a data center

1. **We swipe**
   The swipe, the pause, the rewatch: each is a signal to a server.

2. **The signal goes to a data center**
   Your signals travel to a building full of servers.

3. **An AI model ranks the candidates**
   Chapter 3's recommender scores what you'll watch next.

4. **The next video streams**
   Back to your phone. Repeat for as long as you scroll.

ONE MINUTE OF TIKTOK, ONE TEST

~96 MB

of data moved

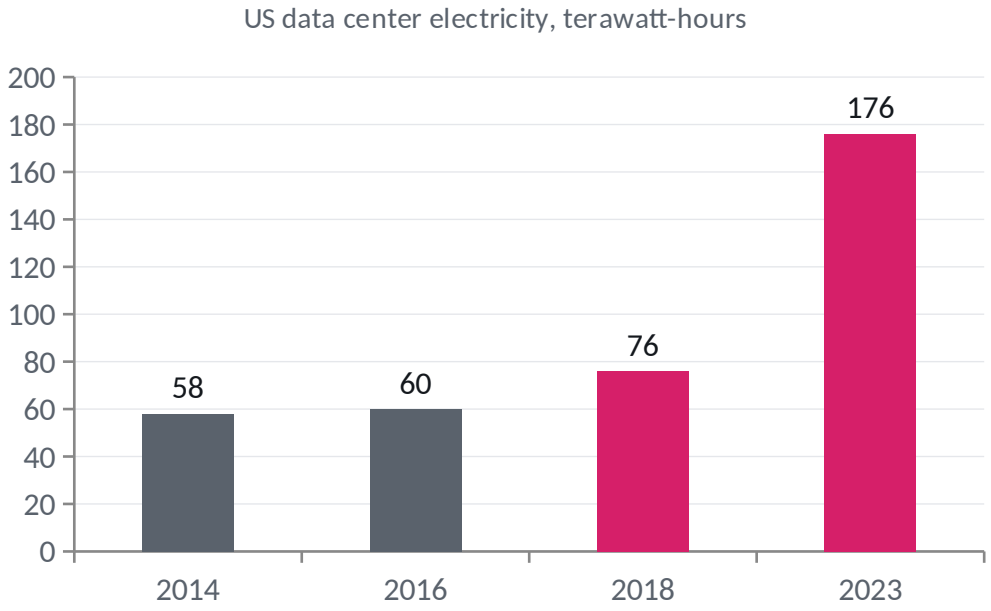
2.63 g

of CO2, counting your phone and the network

Includes your phone and the network, likely most of it. Greenspector, 2021, one phone, one test.

If we scroll a feed, we're using AI, and we were long before anyone typed a prompt into a chatbot.

So, what's been happening over the last 12 years? Data center electricity since 2014. Then AI.



2014 TO 2016

The footprint existed, though.

Streaming, cloud, and feeds exploded. Efficiency gains kept the total roughly flat.

2017 TO 2023

AI broke the balance.

Servers built for AI more than doubled demand. 2023: about 4.4% of US electricity.

Source: Lawrence Berkeley National Laboratory, 2024 data center report (Chapter 5, sections 2.1 and 2.2).

What's an "AI data center," anyway?

1 · A LABEL, NOT A LAW

No official definition

Analysts like the IEA use it for buildings packed with AI chips, wired for far more power, and usually liquid-cooled.

2 · NOT JUST CHATBOTS

LLMs, plus recommendation feeds and ads

From outside, no one knows how much power goes to chatbots vs. feeds. AI overall: about 20% of data center energy today, 40% by 2030 (UN).

3 · AI ISN'T ONLY IN AI BUILDINGS

Everything, side by side

Regular cloud data centers run AI right next to streaming, storage, and email. The building doesn't care what's asking.

4 · NOBODY CAN TELL

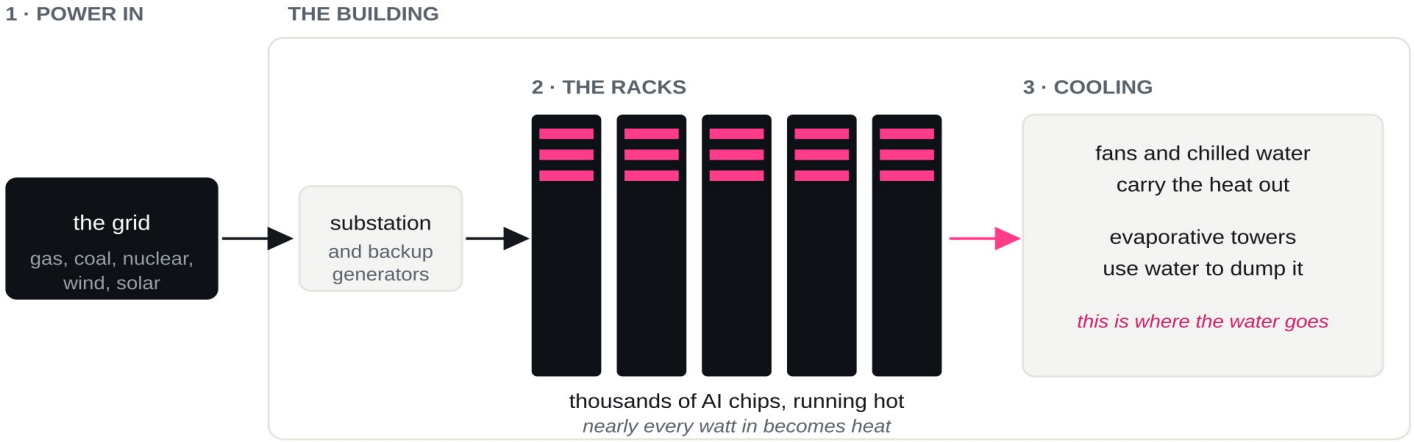
The split is secret

From outside, no one knows how much power goes to which. Rules go by size: the EU's covers every center over 500 kW.

Is it just LLMs? No. An AI data center might be picking your next Instagram Reel.

What happens in the AI data center?

A **data center** is a building full of computers that other people use over the internet. Companies call it “the cloud,” but there is no cloud. There are buildings full of servers, and somebody owns them.



Rough split of the electricity: most of it runs the chips; a smaller share runs cooling and power delivery. All of it ends as heat.

Data Centers: Heat and Water

1. **The computer chips get hot**

   Like a hot laptop, times thousands of chips.

2. **Fans move the heat into the air**

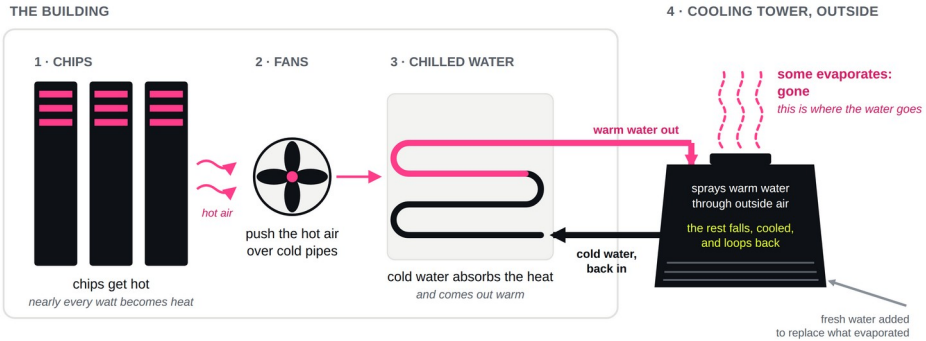
   Air blows across the chips and carries the heat off.

3. **Chilled water absorbs the heat**

   Cold water in pipes pulls heat out of the air. Now the water is warm.

4. **A cooling tower dumps the heat outside**

   Some water evaporates, like sweat. The rest loops back.



The loop water is reused. The water that evaporates in the tower is gone: that's Chapter 5's water cost. A closed loop skips the evaporation and saves water, but its chillers use more electricity.

A laptop fan, the size of a warehouse. The water cost is the part that evaporates.

If data centers already ran TikTok and Netflix, what does AI change?

1 · THE CHIPS

Hungrier chips

AI chips draw two to four times the power of ordinary server chips, thousands to a building, running around the clock.

2 · HEAT AND WATER

Hot chips need more cooling

Evaporating water cools like sweat, and that water is gone. A closed loop saves water but uses more electricity.

3 · STREAM, FEED, OR NEW ANSWER

Not a file, a computation

Netflix fetches a file. TikTok picks a video. ChatGPT writes based on predictions, etc.

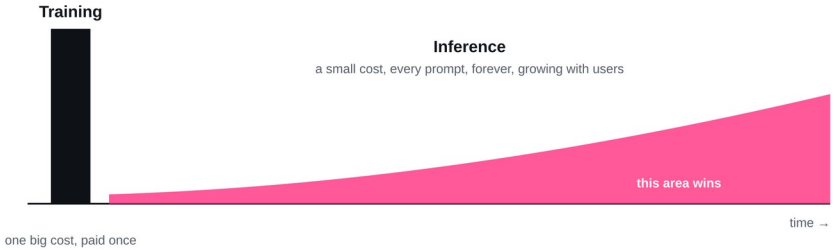
4 · HARDWARE TURNS OVER

62 million tonnes of e-waste

A record in 2022, under a quarter properly recycled. Mostly phones and screens, and faster AI chips push old ones out.

AI didn't invent the footprint. It's the heaviest new work in the building.

Two rounds of environmental impact: *training* and *inference*



TRAINING · ONE TIME, THEN MODEL UPDATE

All AI gets trained, not just chatbots

A company runs thousands of chips for weeks or months to build an AI model (not just LLM). That huge energy bill happens one time, then it's done (until they retrain, of course).

Not from the biggest models. OpenAI, Google, and Anthropic generally don't publish training energy for their newest models. Outside researchers estimate it, and estimates vary a lot.

INFERENCE · EVERY SINGLE USE

A few tenths of a watt-hour

Google: 0.24 Wh and 0.26 mL of water for a median Gemini text prompt.

OpenAI's CEO: about 0.34 Wh.

Epoch AI: about 0.3 Wh.

About a microwave running for one second. Then multiply.

Training impact is less well known. Inference is a bit more known, though spotty.

Small, times billions, times forever

Scale	Energy per day	Carbon	Roughly the same as
You (20 prompts a day)	6 Wh	~3 g	Charging your phone halfway
This class (45 people x 20 prompts per day)	270 Wh	~125 g	One load of laundry, washed cold
All of WWU (about 16,000 x 20 per day)	96 kWh	~45 kg	Three US homes for a day
800 million people, daily, for a year (x 20 per day)	**1.75 TWh**	~820,000 tonnes	**About 160,000 US homes for a year**

Assumes 0.3 Wh a prompt and 20 prompts a person a day (Chapter 5, Table 1). The first three rows are per day; the last is per year; Carbon assumes an average grid, about 0.47 g CO2 per Wh (IEA, 2019 data). Real numbers depend on what the local grid burns. The catch: Google, 2025; IEA, 2026.

THE CATCH

Better per use, bigger in total

Google's median prompt got 33 times more efficient in a year. The IEA says data center electricity still grew 17% in 2025.

Cheaper answers get built into more apps, so total use keeps climbing.

One prompt is tiny. Billions of prompts are not.

So, how do our other tech. habits compare?

LAB 5

Everyday habit	Carbon	About the same as	Source
30 minutes of TikTok	about 79 g	**~550 AI text prompts**	Greenspector
1 hour of YouTube	about 28 g	**~200 AI text prompts**	Greenspector
2 hours of Netflix	about 72 g	**~500 AI text prompts**	IEA
AI specifically	**Energy**	**About the same as**	**Source**
One AI image	about 0.6 Wh	**~30 seconds of streaming (~2 prompts)**	MIT Tech Review
One five-second AI video	about 944 Wh	**~12 hours of streaming (~3,100 prompts)**	MIT Tech Review

One prompt = 0.3 Wh, about 0.14 g CO2 on an average grid (Chapter 5, Table 1; IEA). TikTok, YouTube, and Netflix count your phone and the network; the prompt counts only the data center. Rough, not exact.

READ IT CAREFULLY

Where the energy goes

Streaming: mostly your screen and Wi-Fi. The data center is about 5% (IEA, Netflix). AI: almost all in the data center.

A day of short chatbot questions is small next to an evening of TikTok doomscrolling or Netflix and Chill. A handful of AI videos is not. Streaming's footprint is in your hand. AI's is in someone's backyard.

The AI electricity bill might be landing in someone mailbox who did not agree



An externality: a cost paid by people who were not part of the deal. The data center's owner and its customers get the benefit; the neighbors get part of the bill. Whether they should is a policy question, and section 2.7 is about who gets to answer it.

WHO PAYS MORE · PJM MARKET MONITOR

63 percent, about $9.3 billion

"Data centers raised electricity bills" (PJM's grid watchdog, 2025)
PJM runs the power grid for 13 states and D.C.; its market monitor is the watchdog that checks its prices.

WHO PAID LESS · EPRI, 2015 TO 2024

About 3.5% lower prices

"Data centers lowered bills" (EPRI, an industry-funded group, 2015 to 2024)
EPRI is a research group funded by the power industry.

An externality isn't a side effect. It's a transfer, and the person paying may not have agreed.

Who’s saying all of this?, and how would we know? The numbers are not exact

GOOGLE · PER MEDIAN PROMPT

0.24 Wh | 0.26 mL

A typical Gemini prompt. Counts only Google’s data centers.

WASHINGTON POST · PER AI EMAIL

519 mL

About a bottle. Counts the power plant's water too, for a 2024 model.

MOST FRONTIER MODELS

No number

No company figure at all. Every estimate comes from outside.

WHO BENEFITS FROM THE GAP?

Every side gets something

No number to be held to. A critic can pick the scariest estimate. A utility can forecast whatever justifies its plant.

SOME GOVERNMENTS ARE PUSHING

New Jersey, August 2026

Data centers must report energy and water every six months. That's the building, not the prompt.

Who loses: the public, who can’t check any of them.

The case: from Three Mile Island to Memphis



Three Mile Island, Pennsylvania. Photo: U.S. Department of Energy, public domain.

CLEAN POWER, ON CONTRACT

A reactor comes back

Microsoft is restarting Three Mile Island. Google, Amazon, and Meta back small reactors, none running yet.

WHAT POWERS DATA CENTERS

30% coal / 27% renewables / 26% natural gas / 15% nuclear

Worldwide today (IEA). Nuclear is low-carbon, but not renewable.

MEMPHIS

Gas arrives first

xAI, maker of Grok, ran up to 35 gas turbines where 15 were permitted, in South Memphis. A gas turbine is a jet engine on the ground: fast to set up, and it pollutes the air nearby.

AI companies NEED power, two very different answers.

Elon Musk's Colossus, and coal's second life

XAI · COLOSSUS, MEMPHIS

35 turbines

Gas turbines powering the Grok supercomputer, where 15 were permitted. Extras removed after pushback.

WHAT GAS DOES

Smog

Turbines release nitrogen oxides. UT researchers found they added to Memphis air pollution.

COAL'S SECOND LIFE

20 of 50

US coal plants due to retire by 2035 were delayed or kept open (2026 study, not yet peer reviewed).

WHO LIVES NEXT DOOR

Neighborhoods nobody asked

Colossus sits in mostly Black neighborhoods already ringed by industry. In April 2026 the NAACP sued over xAI turbines in Mississippi.

WHY COAL STAYS ON

Omaha and Georgia

Utilities kept coal units running as server farms grew nearby. Coal is the dirtiest common way to make power.

Nuclear is the headline. Gas and coal are what run while the reactors get built.

Env. regulation for AI? or for all tech?

LAB 5 DEBATE

THE CASE FOR REGULATING EVERY DATA CENTER

Count the building

There's no legal "AI data center." Rules go by building size, like the EU's 500 kW cutoff, not by what runs inside.

Big cloud buildings run AI next to TikTok, and the electric meter can't tell them apart.

Crypto mining used 0.6 to 2.3% of US electricity, and sued rather than reported.

Most **e-waste** isn't AI. The EU already makes every big data center report.

THE CASE FOR STARTING WITH AI

Follow the growth

AI is about 20% of data center energy today, 40% by 2030 (UN). The IEA sees AI-focused centers tripling.

One AI request can cost thousands of times another.

Narrow rules pass. Washington's broad 2026 bill died.

Not one or the other: a floor for every big building, plus extra rules where AI grows fastest.

What could regulation actually be?

1 · REPORT IT

Publish the numbers

Every large data center publishes its energy, water, and e-waste on a schedule. Cost: companies can be held to the numbers.

2 · PRICE IT

Pay your own grid costs

Big data centers pay for their own grid upgrades, not the neighbors. Cost: higher prices, and some may move to states without the rule.

3 · ASK FIRST

Towns get a real say

A vote before the permit, including the right to say no. Cost: slower building, and lost tax revenue if the town says no.

4 · LABEL IT

An energy sticker

Apps and AI models get an energy rating, like a fridge. Cost: a fight over what to measure, and paperwork for every company.

Already real: New Jersey (report it, price it), New York (a one-year pause), the EU (report it).

Let's discuss

1	Regulation would target the data center, but most of our streaming and doomscrolling footprint is our own phones and Wi-Fi, spread across the electrical grid. What's one thing you could change about your own habits to shrink that part?
2	One prompt is tiny; billions aren't. If you could make one rule for data centers (report it, price it, ask first, or label it), which would you pick, and who pays for it?
3	The numbers for calculating the impact of data centers and AI on the environment are a bit murky at the moment. Whose numbers would you trust, and what would you need to check them?
4	Tech companies are betting on nuclear, but the reactors are years away, and natural gas and coal fill the gap until then. Is the push to nuclear a real fix, or a reason to keep burning gas while we wait?

Break About 5 minutes. Then find your project team.

Lab 5 this week, at home

DUE FRI 10/23

1

Read

Articles on AI's energy and water, then streaming, TikTok, and gaming.

2

Build

A green carbon calculator with Claude, compared with a flight, a car, and a burger.

3

Decide

Write a mini-policy for data centers, think through e-waste, and take your debate side.

4

Write

Five short items on Canvas, plus your calculator as an HTML file or PDF.

Due Friday 10/23, 11:59 p.m., with the Chapter 5 quiz. Start early: free Claude has usage limits, so don't leave the build for Friday night.

PART 2 · PROTOTYPE PROJECT · ABOUT 20 MINUTES

Find your group. Start thinking.

Today is just for finding your group of four and getting to know each other. If an idea comes up, great. If not, that's fine too.

Short assignment due Friday, October 30, on Canvas: your team members and your idea. Building is optional, and so is AI.

Today: find your group, start thinking

DUE FRI 10/30

1

Find a group

Teams of three or four. No group yet? Stay put and I'll help you find one.

2

Get to know

Names, majors, and what you're into: jobs, clubs, classes, hobbies.

3

Toss out ideas

Where have you seen a problem worth fixing? Any idea counts at this point.

4

Swap contacts

Set up a group chat so you can keep talking this week.

No idea needed today. Short assignment due Friday 10/30 on Canvas: your team members and your idea. Pitches: three minutes, Shark Tank style, on Wednesday, Dec 2.

The midterm is Wednesday

WED 10/21

WHERE

In our classroom, on Canvas

WHEN

The whole class period, 2:00 to 3:50

WHAT

Multiple choice, plus two short essays

COVERS

Chapters 1 to 4 and Labs 0 to 4

The short essays: pick two of three topics. For each, argue both sides in about 200 to 300 words, using specifics from the chapters and labs, then pick a side.

The five critical AI questions

Q1	**What is the AI system, and what does it do?**	Name it, the company, and the kind: predictive, generative, or agentic. Then the five parts.
Q2	**Why does it exist, and who benefits?**	And who doesn't. Benefits live here.
Q3	**What happens when it fails, and who finds out?**	Risks live here. Loud failures and silent ones.
Q4	**Who's saying that, and how would we know?**	A company's number vs. an outside check.
Q5	**Keep it, fix it, or drop it, and what does that cost?**	Every choice has a price. Name it.

Revisit the three big debates

PRACTICE

1	Is AI going to kill us? Use Lab 1's X player and Harris's four kinds of AI, from tool AI to uncontrollable AI.
2	Therabot: could a chatbot therapist help, or become a quiet failure? The dial, the Narcissus trap, seems vs. is.
3	AI alone, or a doctor in the loop? Every Cure, the ultrasound trial, and the Goh study.
You	Which side would you take now, and what's your best piece of evidence for it?
Do	In your groups, about 10 minutes: make the strongest case for each side, using specifics from the chapters and labs. Then pick one.

Sample multiple choice

PRACTICE

1	A parking garage sign switches to “FULL” when a sensor counts 300 cars inside. Using the course’s test, is it AI? A) Yes, it decides on its own B) Yes, it uses a sensor C) No, it follows a rule a person wrote and never learned from data D) No, it can’t talk
2	A language app’s chatbot tells every user their pronunciation is “amazing,” even when it isn’t. Praised users renew more often. Based on Chapter 2, what’s going on? A) It has started to care B) Someone set the dial toward warmth because it keeps subscribers C) It’s hallucinating D) It’s broken
3	A job site shows you a posting because people who applied to the same jobs you did also applied to it. It never reads the job descriptions. Which method is this? A) content-based filtering B) generative AI C) a hand-written rule D) collaborative filtering
4	A company’s AI writes replies to customer emails. Each one sends automatically unless a manager clicks “cancel” within 10 minutes. Where is the human? A) in the loop B) on the loop C) out of the loop D) there isn’t one
5	A recipe AI scores ingredient pairs from published recipes. It gives miso and blueberry 2 out of 100, but a chef has served it for years. Best reading? A) It tastes bad B) The AI tested it and rejected it C) Few recipes connect them, so the low score means “no data,” not “bad” D) The AI is biased against chefs

Sample answers

PRACTICE

1	C. It follows a rule someone wrote. Nothing was learned from data, so it isn’t AI by our test.
2	B, the dial. The warmth is a setting chosen because it keeps people paying. It seems caring; it isn’t.
3	D, collaborative filtering. It matches you to people like you, not jobs like your jobs.
4	B, on the loop. A person can step in, but the AI acts first. If nobody’s watching, it’s nearly out.
5	C. A low score can mean “nothing connects,” not “bad.” Same as the sparse dots in Lab 4’s graph.

PART 3 · MIDTERM REVIEW · ABOUT 30 MINUTES

Does anyone have any questions about the midterm?

The midterm doesn't ask you to memorize every number. It asks whether you can look at an AI system you've never seen and run the five critical AI questions on it.

ATTENDANCE BEFORE 4PM PLEASE!

Attendance = your notes from today, uploaded to Canvas right after class.

Wednesday 10/21	The midterm, in our classroom, on Canvas. Chapters 1 to 4, Labs 0 to 4.
This week, at home	Lab 5, The Footprint Machine. Start early. And meet with your project team.
Friday 10/23, 11:59 p.m.	Lab 5 writeup with your calculator (HTML file or PDF), and the Chapter 5 quiz. Next Friday, 10/30: your team members and your project idea.

Tonight: scroll your feed, and remember where it lives.